Group Assignment 2: Exchange Rate Forecast

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Data

We analyze the Japanese yen relative to the U.S. dollar and quote the exchange rate as

$$S_0 \equiv E_{\mathrm{JPY}/\$} = \text{yen per 1 U.S. dollar.}$$

We use the spot and forward quotes:

$$S_0 = 152.904 \text{ JPY}/\$.$$

The interest rates are annualized government yields with matching maturities in each country (3-month and 1-year):

- U.S. 3-month interest rate: $i_{\$}^{3\mathrm{mo}} = 3.69\%$
- Japan 3-month interest rate: $i_{\mathrm{JPY}}^{3\mathrm{mo}} = 0.72\%$
- U.S. 1-year interest rate: $i_{\$}^{1\mathrm{yr}} = 3.44\%$
- Japan 1-year interest rate: $i_{\mathrm{JPY}}^{1\mathrm{yr}} = 1.05\%$

Interest Rate Parity

Covered interest rate parity:

$$\frac{1+i_{\$}}{1+i_F} = \frac{F_F^{\$}}{E_F^{\$}}.$$

The forward rate is:

$$F^{\mathrm{IRP}} = S_0\left(\frac{1+i_{\mathrm{JPY}}}{1+i_{\$}}\right).$$

Uncovered interest rate parity is:

$$\frac{1+i_{\$}}{1+i_F} = \frac{E^{1\mathrm{yr}}{}_F^{\$}}{E_F^{\$}}.$$

(1) IRP Forecasts

3-month implied forward rate

$$F_{3\text{mo}}^{\text{IRP}} = 152.904\left(\frac{1+0.0072}{1+0.0369}\right) = 148.519 \text{ JPY/\$} \approx 148.52.$$

1-year implied forward rate

$$F_{1\text{yr}}^{\text{IRP}} = 152.904\left(\frac{1+0.0105}{1+0.0344}\right) = 149.371 \text{ JPY/\$} \approx 149.37.$$

(2) Market Forward Rates and Comparison

From the USD/JPY forward-rates from FXEmpire we use:

$$F_{3\text{mo}}^{\text{mkt}} = 151.758, \qquad F_{1\text{yr}}^{\text{mkt}} = 148.998.$$

Horizon	IRP forward F^{IRP}	Market forward F^{mkt}	Difference
3 months	148.519	151.758	−3.239
1 year	149.371	148.998	0.373

Discussion

Since U.S. interest rates exceed Japanese interest rates at both maturities ($i_{\$} > i_{\text{JPY}}$), Covered IRP implies that the yen should trade at a forward premium relative to the dollar, which means the forward rate should be below the spot rate. This is what we observe for the 1-year forward rate.

Small differences between the IRP-implied forward and observed market forward rates can arise from bid-ask spreads, timing mismatches across sources, and differences in quote/compounding conventions for interest rates and forward contracts.

Sources

- Spot rate (USD/JPY): Yahoo Finance
  `https://finance.yahoo.com/quote/JPY%3DX/`
- U.S. 3-month rate: CNBC
  `https://www.cnbc.com/quotes/US3M`
- Japan 3-month rate: Investing.com
  `https://www.investing.com/rates-bonds/japan-3-month-bond-yield`

- U.S. 1-year rate: CNBC
  `https://www.cnbc.com/quotes/US1Y`
- Japan 1-year rate: Investing.com
  `https://www.investing.com/rates-bonds/japan-1-year-bond-yield`
- Spot and forward rates (Bid/Ask/Mid): FXEmpire
  `https://www.fxempire.com/currencies/usd-jpy/forward-rates`